

Detection Techniques of DDoS Attacks:A Survey

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Abstract— Security is the main confront of internet, Distributed Denial of Service (DDoS) is the major cause of threat. DDoS reduces the network resources, and results in bandwidth depletion. The main aim of DDoS is to prevent legitimate users from accessing the services. There also exists a difficulty to differentiate between flash crowd and DDoS attack traffic. The various existing solutions which are given in order to detect DDoS has been discussed in this paper. The goal of the paper is to review how the different methods are helping in downsizing the effect of DDoS and also how that are being used to detect a DDoS attack.

Keywords— Distributed Denial of Service (DDoS); Detection; Flash Crowd; Traceback; Entropy; Intrusion Detection System(IDS); Intrusion Prevention System(IPS)

I. INTRODUCTION

Even though the world has changed a lot during the past 10 years due to the rapid development of technology. There has been an internet facility for all due to changes in computing and needs of the users. Nowadays, every office, home is connected to the internet with a broadband. So to access the resources and wide area networks has become easy due to dynamic and easy to access internet facility. But the traditional networking principles are not changed, they are built using the switches and routers which are still being manually configured by the operators. So, it arises some of the risks such as traffic flows due to traffic passing across the network devices, interfaces and hosts, vulnerabilities arises due to unupdated patches in network devices[2] [28] [29]. There are also chances of attacks on admin servers due to unsecure servers or lack of security and also the single point of failure due to unreliable controllers.

Out of the various challenges, Security is the main challenge and concern in the internet environment. Some of the common threats are access control attacks, injection and execution of malicious software, sniffing, and masquerading [1]. One of the main threat to security is Distributed Denial of Service attack (DDoS) which is the most unpredictable. DDoS is a fast spreading problem in today's networking environment which are still using the old technique of manually configuring the network topology [26] [27]. In Denial of Service (DOS) attack a single source performs the attack whereas DDoS uses multiple hosts to attack against a system. It is very difficult to identify the attack source when such an attack occurs as the attacker's hides their identities by spoofing their IP address. Example North Korea of DDoS attacks on South Korea. According to the survey carried out by Arbor [23] networks in 2008 [24], the DDoS attacks are growing gradually since 2001.

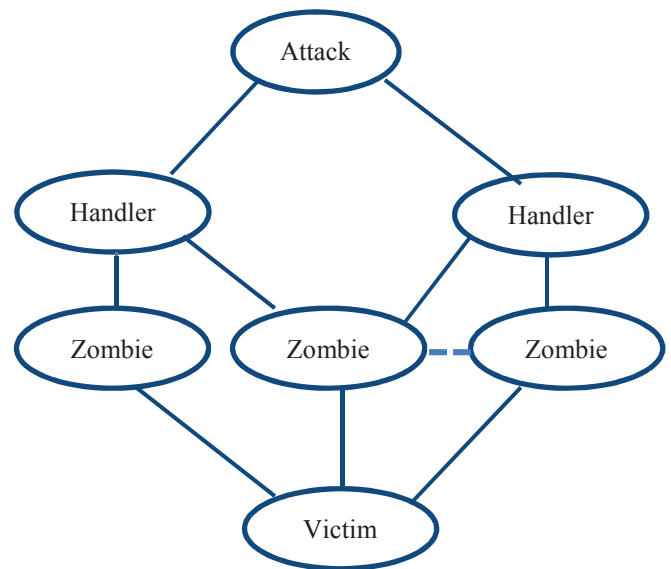


Fig. 1. Architecture of DDoS Attack

Approximately 1200 DDoS attacks occur per day.

The aim of the attacker is to discover the vulnerable sites or hosts in the network, which are running no antivirus code or outdated antivirus code. They first establish a network by the use of systems which are called as zombies, which is being controlled by botmasters. Then they try to install new programs on the compromised hosts on the network. These compromised hosts are known as bots or zombies which send useless or bogus packets in order to disrupt the victim. DDoS frazzle the victim and their CPU resources, by triggering errors within the firmware of machine. The aim behind the attack of the attacker is to prevent the legitimate users to access the resources. The companies which has been pruned by DDoS has reported by the theft of funds, data and intellectual property. In 2013, [3] [23] [24] DDoS has incapacitate companies websites leading to the downtime which are resulting in closure of business operations and hobbling their brand images.

The companies scent that the problem is not going away and believe that the threat is undiminished. According to the survey carried out by Arbor Networks in 2013, the DDoS attacks has increased over 8x, whereas the no of attacks over 20 Gbps since 2012. Approximately 1200 DDoS attacks occur per day. They are now consuming more of the manpower to detect them. It is broadly classified in two types one is direct attack and other is reflector attack. During the time of direct attack the attacker

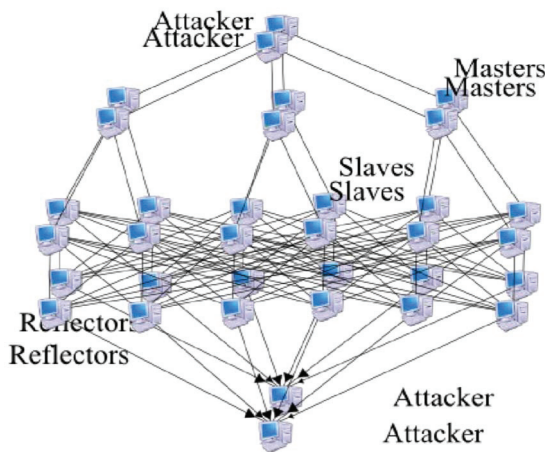


Fig. 2. Architecture of DRDoS Attack

takes effort to send immense amount of attack packets to the victim. The attack packets can be in any form such as Transport Layer Protocol (TCP), UDP, and Internet Control Message Protocol (ICMP) etc. In a reflector attack is an indirect attack which uses intermediary nodes (routers and servers) known as reflectors which are used to launch the attack on the victim. It is known as Distributed Reflection Denial of Service attacks (DRDoS). It is a difficult problem to identify the actual starting point of packet due to the IP spoofing (IP address is forged) as the internet is dynamic, stateless, anonymous in nature [10].

The stateless nature of the network topology means that there are no record of the previous interactions. Each packet travels with its own without reference of any other packet.

Due to similar behavior of the legitimate network events as flash crowds, it is difficult to discriminate between flash crowds and DDoS. When large no. of users tries to access a particular website is known as a flash crowd. Example Online ticket booking, Online Share Trading. So the detectors mistreat legitimate flash crowds as DDoS traffic. The high-profile servers such as banks, social networks, credit cards payment gateways are main target during attacks. In 2010, attack occurred on Twitter [23], server remained down for few hours. The application level [4] methods has the capability of differentiating legitimate flash traffic. DDoS which aims at the network layer such as ICMP flooding, SYN flooding, UDP flooding are known as NET-DDoS attack whereas at application layer are known as Application DDoS [4].

- Non-Application layer method attacks- They make use of spoofed IP address. The botnet attacks ensconce their IP address by spoofing a valid IP address but the zombie attacks did not hide their IP address [4]. The Authors proposed the algorithm such as puzzle solving, pushback blocking, source based filtering, monitor TTL (time to live) to differentiate flash traffic from attack traffic. The flash traffic is coming to the site for different reasons and retains the different statistical properties due to which it is misinterpreted as

attack traffic [17]. So in order to detect and block the DDoS Intrusion Detection System [18] and databases of attacks signatures are used [21].

- The approach used by authors were statistical approach but fails to screen flash traffic from attack traffic. Some of the other approaches used were CAPTCHA, filters, Hidden Semi Markov Model to monitor DDoS attacks. But the fragility arises when new legitimate traffic possess the identical statistics as archaic traffic.

II. DDOS ATTACKS

In this era, there are various types of DDoS that are being used in the networking environment in order to jounce the company's website by hampering their operations. They are-

A. SYN Flooding

SYN Flooding is a type of DDoS attack which works by deeding the weaknesses on TCP. The attacker sends SYN packets to ports-state of 'Listening' in target host. Usually, it contains source address of the actual host, but during the attack source address does not depicts actual host. When the target receives the SYN packets from host, the target retorts by sending SYN/ACK packets. It waits for ACK from the host in order to replete the connection process. As the source address sent in the SYN packet, by the attacker is invalid, ACK will never be acquired by the target, and the target of attack ports wait till the connection times out.

B. ICMP Flooding

ICMP flooding escapade the configuration errors in network devices. The attacker consigns a immense no of ICMP packets with bogus return address to the server so that the server does not have the time to retort to other servers. It does not interchange the data between the systems. The Network bandwidth is bled, due to which legitimate packets are repudiated by the servers. In this, the attacker spurt a controller program (Trojan horse) on master computer, who then trains the slave computer to spread attack on victim.

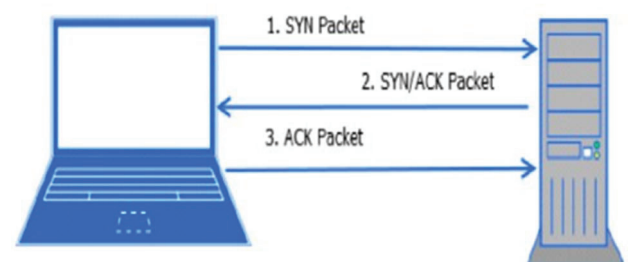


Fig. 3. SYN Flooding

C. UDP Flooding

This type of DDoS attack uses User Datagram Protocol which is a connectionless and sessionless network protocol. It does not binds a three-way handshake like TCP and it runs with a lessened overhead so it could be purged with minor resources

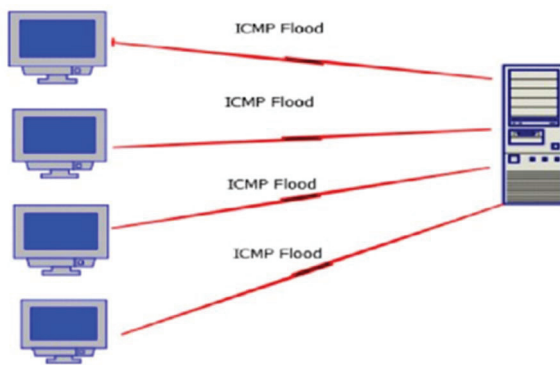


Fig. 4. ICMP Flooding

in hand. In this attacker deluge the arbitrary ports on the targeted host with IP packets containing the UDP datagram. The receiving hosts review on applications with these datagrams, when there is no correlated application is listening it response with an ICMP destination unreachable packet. Some of the UDP flood attacks can take the form of DNS Amplification attacks. Both amplified and non-amplified UDP can be emanated from botnet cluster of various sizes. There are a no of commercially accessible software packages used to perform a UDP Flood attack (ex. UDP Unicorn).

III. EXISTING SOLUTIONS

The count measures of DDoS consists of three components detection, defence (or mitigation) [21], IP traceback. Detection is foremost and a crucial step in any of the attack mechanism. The attackers can smoothly moron the detection algorithms [21] because of the open architecture in the internet, attackers can easily deceit the source IP address and also undermines the Source Address Based Detection algorithm. They can also transforms the Time To Live(TTL) value of the packets which have been attacked by bluffing the Hop-Count Detection method [21].The main question which arises is how the traffic can be blocked without affecting legitimate users. As internet is an open and anarchic platform, so it is rigid to conduct operations in a generous scale.

The Motivation factor which gave surge to solution for DDoS are

1. Revenue loss
2. Invariably slow network performance
3. Service unavailability

In the network environment, routers can work to conduct DDoS attack detection [5] [21] and attack packet filtering [25] to avoid calamitous damage. Ancient packet marking techniques are no longer applied because of high memory consumption and poor measurability. So the use of Entropy Variation helps to extricate between traditional and DDoS attack traffic. Some of the extant methods which are used in ongoing network environment for detection and mitigation of DDoS [21] attacks are as follows.

A. Traceback methods

The information processing traceback means that the predilection of the peculiar supply of any packet sent across the internet [6].Currently due to the vulnerability of the present architecture of the internet, we do not have the tendency to recognize the attackers .In fact, information processing traceback schemes are reflected if they will able to detect DDoS attack as the packets arrives in internet. So the subsequent methods are as follows.

1) IP Traceback Technique-

It is an integral step in defence against DDoS attacks using IP spoofing [10].They determines the definite source or origin of packets which are transmitted over the internet [6].The major difficulty arises in detecting source of an attack is due to inaccurate or deceived IP address [8] .The attack is performed by the attacker from the mutiple sources.The IP traceback helps to mitigate the consequence of the attack and to filter out the suspicious traffic [6][8].They are widely classified into two schemes one is proactive and another is reactive.

- Reactive approach- The IP traceback is conducted in this approach, as the detection of attack [21] are possessed. It must be outright when the attack is active and becomes ineffectual as it ceases [8].It is segregated into IDS and Non-IDS assisted. IDS assisted schemes are further classified into Network Based and Host Based. Example are Input debugging and controlled flooding.
- Proactive approach-In this approach, the information is traced in the form of the packets as they passes in the network [8]. The victim make use of the resulting traceback data for identification of attacker and reconstructs the attack path [9].It is also restricted as Out-of-Band and In-Band information. Example Logging, Messaging, Packet Marking [16].

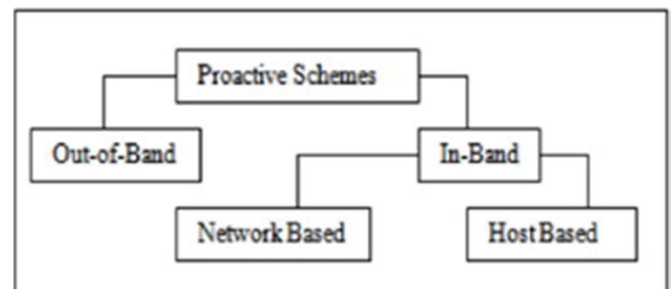


Fig. 5. Topology of Proactive Classification

2) Packet Marking Techinque

The Traceback methods are backed by packet marking .The main techinques which are used to mark the packets are PPM and DPM [6]. The techniques are discussd as follows-

- Probabilistic packet marking(PPM)- In this technique the packets are being marked by the routers as they cross them [16].It is amenable to the attackers as

attackers will send deceived marking information in order to misguide the victim [6]. The victim then reconstruct the way attack packets went through. It is done in two ways* Node marking-It uses the router IP address. Edge marking-It uses the edges of paths.

PPM requires large no of packets, so there is an high computational load [8]. The drawback of this technique are [16] - It suffers from space problem to store great amount of marked packets to reconstruct attack diagram, Large no of false positives, Slow convergence rate, Only for single attacker, It results in 95% success, when there is a need of 300, 00 packets for reconstruction.

- Deterministic packet marking (DPM)-In this technique every incoming packet is marked by the router with a unique identifier. The spoofed packets are overwritten with correct marks. It require less overhead and computation than PPM [6]. It converges very quickly and it has zero false positive. A downside of this scheme is some of the packets will not be overwritten by any routers [8]. Attacker can also write some irrelevant information in all the packets knowingly in order to confuse the victim. Existing traceback mechanism suffer from drawbacks- High process and storage costs, Little scalability to high attacker populations, Poor performance is presence of legacy routers.

B. Entropy Variation

Entropy is of times used to detect DDoS attacks productively, which represents the random feature of network traffic. It has been used in anomaly detection [21]. It depicts the concentration and disseminated characteristic of traffic. But the entropy depends only on the values computed by each packet field. In DDoS Attack, feature distribution will become more scattered, as when source addresses are spoofed. So the entropy variations are used to trace back to the source of the DDoS attacks.

The use of entropy variation discriminates between traditional and DDoS attack traffic. The entropy variation is defined as the change in randomness of flows on the router. Entropy of flow metric is used for attack pattern independent detection [21]. It uses divergent metrics for normal variation or high order moments of flow [7]. Once victim recognizes progress attack, it will push back to the Local Area Networks (LANS), by supporting data of flow entropy variations, and is able to segregate location of attackers. The leading flows which are causing dropping of flows entropy on routers are identified and are treated as suspicious flow. Flow entropy on LAN routers are used to supervise network traffic and it raises alarm when the entropy level drops significantly.

The information distance is a term which is used to discriminate DDoS attacks from flash crowds. If the information distance among suspicious flows are less than a given threshold is depicts it as DDoS attack otherwise it is a flash crowd. The relative entropy is also not considered as an excellent metric [7], as it introduces false positive or false negative.

The Entropy can be calculated such as

$$H(Y) = -\sum \Pr[Y=y] \cdot \log(\Pr[Y=y])$$

Where y is sample space of Y, it measures uncertainty of Y.

The flow entropy is stable in non-DDoS attack whereas it decreases in DDoS attack. The detecting algorithm is made to run on entire routers in Local Area Network (LAN). The flow entropy is used as a metric especially for the edge routers in order to monitor the network traffic. In the absence of attack, the flow entropy rests in a uniform range. If information distance is less than a given threshold, then the DDoS attack is confirmed and routers start discarding packets from the cautious flows before they reach the victim.

C. Intrusion Detection and Prevention System(IDS/IPS)

A DDoS attack prohibits the legitimate users from assessing the services. One of the most competent mechanism of detecting and prevention [19] of DDoS[13] is Intrusion Detection System(IDS) [12]. It is a system in the form of a application which usually monitors the network for suspicious events and generates a report to the administrator in order to take an action against it[5] [18]. Traditional IDS/IPS techniques such as signature based detection, anomaly detection etc. [5] [13].

1) Signature Based Detection-

The signature Based intrusion detection[18] intends to define a set of rules or signatures or predefined knowledge base that can be used to decide that a given pattern is that of an intruder [12]. The one of the reason to use it is serenity in maintaining and updating preconfigured rules in the database.

A signature is generated for each user that determines whether user is suspicious or not. Captcha is used for determination of signature whether represents attack whether their is a non-human user or legitimate user. If the load on server is low, then no blocking is done. But if the load on server goes above LLT(low load threshold), then suspicious users are detected and delayed. If it reaches HLT(high level threshold) then suspicious users are deterred [12]. It uses signature detection for detecting suspicious users. Traffic with similar signature are valid, not blocked or delayed. But in case of repeated failure of signature type, the users are delayed or blocked.

2) Anomaly Detection-

Anomaly(or behavioral) detection is concerned with identifying events that appear to be anomalous [15] with respect to normal system behavior. The techniques such as data mining, statistical modeling and hidden markov models are used [12]. It involves the collection of data relating to the behaviour of legitimate users over a period of time, and then apply statistical tests to the observer behavior, which determines whether that behavior is legitimate or not. It helps to generate the rules in such a way that can lower the false alarm rate for unknown as well as known attacks.

IDS can be classified into Host based IDS(HIDS), Network based IDS (NIDS), Distributed IDS(DIDS) and Hybrid based

IDS.IDPS essentially focuses on analyzing the viable incidents,gathering the information regarding them,and generating reports.It responds to a detected threat by attempting it to not succeed further.It uses various techniques which involve stopping attack itself,changing attack environment by help of reconfiguring a firewall.

IV. CONCLUSION

The review paper, mainly characterizes the DDoS attack ,different types of DDoS attacks that can occur.DDoS is a very briskly emerging problem.It also deals with the various different existing solutions which helps in the detection of DDoS attack such as traceback mechanisms, using IP traceback classified into proactive and reactive approach, packet marking such as PPM and DPM, and the entropy variation.Intrusion Detection[18] and Prevention System [19]is also one of the way which helps in the detection.The various existing techniques has their own advantages and disadvantages.

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